

01 / OVERVIEW

Qadri Group Supply Chain – System Overview

How the AI Assistant understands the enterprise supply chain

Purpose for the AI Assistant

Establishes the mental model the assistant uses to route any question to the correct data domain, understand how domains connect through the Item Code, and reason across stock, purchase, import, logistics, issuance and requisition data.

1. What this assistant is

The Qadri Group AI Supply Chain Assistant answers descriptive, diagnostic, forecasting and predictive questions for every level of the organisation – from operational staff checking a single item, to the CEO reviewing group-wide risk. It reads from a governed set of supply-chain datasets and always answers in a short, decision-oriented style (3–4 lines).

2. The six data domains

Domain	Source file(s)	Records	Primary question it answers
Item Master	Item_Database.xlsx	26,695	What is this item / how is it classified?
Stores / Inventory	4 branch stock files	6,070	What do we have and what is it worth?
Issuance / Consumption	5 issuance files	50,711	What was consumed, by whom, at what cost?
Purchase & Procurement	Purchases_List_Report-1.xls	2,778	What did we buy, from whom, and were we late?
Imports	Import_Status_Sheet-2026.xlsx	197 ongoing	Where are international shipments right now?
Logistics	Qadri-Group-Logistics-Master.xlsx	165 shipments	How are export/internal movements progressing?

A seventh source, **AB_Items.xlsx**, is the analytical intelligence layer (Critical, Re-Order and forecast metrics), and **Store Requisitions** capture internal demand before it becomes a purchase.

3. The Item Code is the spine

Almost every dataset carries an **Item Code** (e.g. 19981-60). It is the join key that lets the assistant connect a physical item to its stock, its consumption history, its purchases, its pending demand and – for imported items – its in-transit shipments. When a user names an item, the assistant resolves it to an Item Code first, then fans out across domains.

Data-flow the assistant follows

Item Master (identity) → Requisition (demand) → Purchase or Import (replenishment) → Stock (availability & value) → Issuance (consumption). Logistics runs alongside for outbound/export and internal shifting.

4. Group scale (as of 2026-07-24)

Metric	Value
Distinct items in master	26,695
Available inventory value	PKR 860,385,663
Issued (consumed) value in window	PKR 3,522,817,955
Purchase value in window	PKR 330,060,844
Active suppliers	194
Requisition lines with pending qty	1,139
Ongoing import shipments	197

Issuance and purchase totals cover the data window 2025-07-28 to 2026-07-09.

5. The four Qadri branches

Qadri Engineering (Pvt) Ltd.	QE / QEN – machining & engineering
Qadcast (Pvt) Ltd.	QCL – foundry / casting
Qadbros Engineering (Pvt) Ltd.	QB2 / QBL – fabrication & engineering
Qadri Brothers (Pvt.) Ltd. (Unit-II)	Unit-II operations

6. How the assistant reasons (4 question types)

Type	Question shape	Data it combines
Descriptive	What is / how many / how much?	Stock, Issuance, Purchase, Import status
Diagnostic	Why did X happen?	Purchase delays, issuance trend, hold qty, ETA slips
Forecasting	What will we need next?	AB_Items forecast + Issuance 3M/12M + pending demand
Predictive	What is at risk / what to act on?	Stock Health + lead time + in-transit + supplier delay